

PART II

PROJECTS

Congratulations! You now know enough about Python to start building interactive and meaningful projects. Creating your own projects will teach you new skills and solidify your understanding of the concepts introduced in Part I.

Part II contains three kinds of projects, and you can choose to do any or all of these projects in whichever order you like. Here's a brief description of each project to help you decide which to dig into first.

Alien Invasion: Making a Game with Python

In the Alien Invasion project (**Chapters 12, 13, and 14**), you'll use the Pygame package to develop a 2D game. The goal of the game is to shoot down a fleet of aliens as they drop down the screen, in levels that increase in speed and difficulty. At the end of the project, you'll have learned skills that will enable you to develop your own 2D games in Pygame.

Data Visualization

The Data Visualization projects start in **Chapter 15**, where you'll learn to generate data and create a series of functional and beautiful visualizations of that data using Matplotlib and Plotly. **Chapter 16** teaches you to access data from online sources and feed it into a visualization package to create plots of weather data and a map of global earthquake activity. Finally, **Chapter 17** shows you how to write a program to automatically download